

3. Shuttle Management System: A Smart Campus Transit Solution.

The **Shuttle Management System** is designed to provide **efficient, cost-effective, and seamless** transportation for students within a university campus. The system streamlines shuttle bookings, trip tracking, and driver availability management, ensuring a hassle-free commuting experience.

I. Shuttle Booking Management

- **Add, View, and Edit Bookings:**
 - Users (students or staff) can book shuttle rides through the system.
 - Admins can view and modify existing bookings if needed.
- **Trip History Tracking:**
 - Users can view their past trips, including details like date, time, route, and driver information.

II. Driver Availability Management

- **Driver Schedule Timeline View:**
 - A visual timeline displaying the availability of drivers for a given day.
 - Ensures admins can efficiently allocate drivers based on demand.
- **Hourly Scheduling Controls:**
 - Admins can add, view, and modify a driver's **start duty, end duty, and breaks** within an **hourly time frame**.
 - Helps optimize driver shifts and reduces idle time.

III. Admin Functionalities

- **Manage Shuttle Routes:**
 - Define and update shuttle routes, including pickup/drop-off points.
- **Assign Drivers to Routes:**
 - Ensure that each shuttle trip has an available driver.
- **Monitor Shuttle Usage & Demand:**
 - Track peak hours and optimize shuttle schedules accordingly.

IV. Benefits & Impact

- **Enhanced Convenience:** Students can seamlessly book rides and track shuttle availability.
- **Optimized Driver Scheduling:** Reduces idle time and ensures efficient workforce management.

- **Cost-Effective Operations:** Balances demand with available drivers, reducing unnecessary trips.

Screenshots

Driver journey view and edit:

Driver Management

Dec 16, 2021

Search driver

Samuel Jones

Online

Bob Jones

Offline

Jonathan Spikes

Offline

Start Duty

End Duty

Add Break

2 Pickup

3 Drop

Duty Start

Duty End

Pickup/Drop

Break

Vehicle Change

Empty Leg

Booking Management

Dec 16, 2024

Search Emp, ID, Booking ID

Booking ID	Employee	Status	From	To	Vehicle	Requested Pickup Time	Pickup Time	Planned Drop	Actual Drop	
123123	Thompson	Accepted	Library	Data Centre	NB-002-RF	11:21	-	11:32	11:32	View
324235	Daniel Radcliff	Waiting	Library	Parking	NB-002-RF	11:34	-	11:43	11:43	View
545232	W.J. Smith	No Show	Data Centre	Parking	NB-002-RF	11:50	-	12:10	-	View
434532	Tina Shah	Declined	Library	Data Centre	-	11:55	-	-	-	View
545232	W.J. Smith	Completed	Data Centre	Parking	NB-002-RF	11:58	12:25	12:35	12:40	View
434532	Tina Shah	Requested	Library	Data Centre	NB-002-RF	11:55	-	12:15	-	View
545232	W.J. Smith	On Going	Data Centre	Parking	-	11:58	12:25	12:35	-	View
434532	Tina Shah	Cancelled	Library	Data Centre	-	11:55	-	12:15	-	View
545232	W.J. Smith	Dropped	Data Centre	Parking	NB-002-RF	11:58	12:25	12:35	12:40	View
434532	Tina Shah	Declined	Library	Data Centre	-	11:55	-	-	-	View

Showing 1-10 of 50 items

1

2

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5

>

Employee journey view and edit:

Move
inSync

TrackingPerformanceManagement

Driver Management

Search driver

Samuel Jones
Online

Bob Jones
Offline

Jonathan Spikes
Offline

6:007:008:009:0010:0011:0012:0013:0014:0015:0016:00

-8

9

10

11

12

13

14

15

16

-8

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-8

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16

Duty Start

Duty End

Booking Management

Booking IDEmployeeStatusFromToVehicleRequested Pickup TimePickup Time

123123ThompsonAcceptedLibraryData CentreNB-002-RF11:21--

324235Daniel RadcliffWaitingLibraryParkingNB-002-RF11:34--

545232W.J. SmithNo ShowData CentreParkingNB-002-RF11:50--

434532Tina StuhDeclinedLibraryData Centre--11:55--

545232W.J. SmithCompletedData CentreParkingNB-002-RF11:5812:25

Booking ID: 123123

Thompson

Emp ID: 123123

Waiting

Sign in:

Tue, Dec 17

NB-002-RF

UA3282

White Bus

12

Library

Requested Pickup Time: 11:21

11:25 (+5)

Data Centre

Planned Drop: 11:32

Steve Smith

+1-323-493-3299

4.5

Sign in rider

Mark rider as No-show

Cancel Booking

Edit